

$$13. (4) \quad y'' = 2(y'' - 1) \cot x \quad \text{令 } y' = z, \quad y'' = z'$$

$$z' = 2(z - 1) \cot x$$

$$\frac{dz}{dx} = 2(z - 1) \cot x$$

$$\textcircled{1} z \neq 1 \text{ 时, } \frac{dz}{2(z-1)} = \cot x \cdot dx$$

$$\frac{1}{2} \ln|z-1| = \ln|\sin x| + C$$

$$z - 1 = C \sin^2 x \quad \text{则 } y' = C \sin^2 x + 1$$

$$y' = C_1 \left(\frac{1}{2} x - \frac{1}{4} \sin 2x \right) + x + C_2$$

$$y = C_1 \left(\frac{1}{4} x^2 + \frac{1}{8} \cos 2x \right) + \frac{1}{2} x^2 + C_2 x + C_3$$

② $z = 1$ 时, 亦满足题意,

$$y'' = 1, \quad y' = x + C_1, \quad y = \frac{1}{2} x^2 + C_1 x + C_2$$

可以计入情况①中

$$\text{综上, } y = C_1 \left(\frac{1}{4} x^2 + \frac{1}{8} \cos 2x \right) + \frac{1}{2} x^2 + C_2 x + C_3$$

$$14. (4) \quad (1 + x\sqrt{x^2 + y^2}) dx + (-1 + \sqrt{x^2 + y^2}) y dy = 0$$

$$\frac{\partial Q}{\partial x} = \frac{xy}{\sqrt{x^2 + y^2}} = \frac{\partial P}{\partial y} = \frac{xy}{\sqrt{x^2 + y^2}}, \text{ 是全微分方程}$$

$$\int (1 + x\sqrt{x^2 + y^2}) dx = x + \frac{1}{3} (x^2 + y^2)^{\frac{3}{2}} + C_1$$

$$f(x, y) = x + \frac{1}{3} (x^2 + y^2)^{\frac{3}{2}} + \varphi(y)$$

$$\frac{\partial f}{\partial y} = \frac{1}{2} \cdot 2y \cdot \sqrt{x^2 + y^2} + \varphi'(y) = -y + y \cdot \sqrt{x^2 + y^2}$$

$$\therefore \varphi'(y) = -y, \quad \varphi(y) = -\frac{1}{2} y^2 + C$$

$$\therefore f(x, y) = x + \frac{1}{3} (x^2 + y^2)^{\frac{3}{2}} - \frac{1}{2} y^2 + C$$

$$15. (5) (x dy + y dx) \sqrt{1-y^2} + xy dy = 0 \quad (xy \neq 0, y \neq \pm 1)$$

$$\sqrt{1-y^2} dx + xy dy = 0$$

$$\frac{1}{xy} dx + \frac{1}{\sqrt{1-y^2}} dy = 0$$

$$d \ln |xy| + d \arcsin y = 0$$

$$\therefore \ln |xy| + \arcsin y = C$$

$$16. (1) x dx = (2xy^3 dx + 3x^2 y^2 dy) \sqrt{1+x^2}$$

$$x dx = \sqrt{1+x^2} \cdot d(x^2 y^3)$$

$$\therefore d\sqrt{1+x^2} = d(x^2 y^3)$$

$$\therefore \sqrt{1+x^2} - x^2 y^3 = C.$$

$$16) e^x dx + (e^x \cot y + 2y \cos y) dy = 0$$

$$\frac{\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}}{P} = \frac{e^x \cot y}{e^x} = \cot y$$

$$\therefore e^{\int \cot y dy} = \sin y \text{ 是一个积分因子.}$$

$$e^x \sin y dx + (e^x \cos y + 2y \cos y \sin y) dy = 0$$

$$\int e^x \sin y dx = e^x \sin y.$$

$$\therefore \frac{\partial f}{\partial y} = e^x \cos y + \varphi'(y) = e^x \cos y + 2y \sin y \cos y$$

$$\varphi'(y) = 2y \sin y \cos y = y \sin 2y$$

$$\begin{aligned} \int y \sin 2y dy &= -\frac{1}{2} \int y d(\cos 2y) = -\frac{1}{2} (y \cos 2y - \int \cos 2y dy) = -\frac{1}{2} (y \cos 2y - \frac{1}{2} \sin 2y) \\ &= \frac{1}{4} \sin 2y - \frac{1}{2} y \cos 2y \end{aligned}$$

$$\therefore f(x, y) = e^x \cos y + \frac{1}{4} \sin 2y - \frac{1}{2} y \cos 2y + C$$

9.3

2. 下列函数在相应的闭区域上是否满足利普希茨条件?

① $f(x, y) = \sqrt{y}$, $R: |x| \leq a, 0 \leq y \leq b$;

(2) $f(x, y) = \sqrt{y}$, $R: |x| \leq a, 0 < \delta \leq y \leq b$;

(3) $f(x, y) = xy^n$ (n 为正整数), $R: |x| \leq a, |y| \leq b$;

(4) $f(x, y) = xy^2$, $D: |x| \leq a, -\infty < y < +\infty$.

③. 我们将本节中的存在和唯一性定理应用于下列情况: $f(x, y) = x^2 y + x, x_0 = 0, y_0 = 0, R = \{(x, y) \mid |x| \leq 1, |y| \leq 1\}$. 试估计利普希茨常数 L 及 $M = \max_R |f(x, y)|$ 的值, 并给出定理中解存在的区间长度 $2h$ 的估计值.

$$2. (1) f(x, y + \Delta y) = f(x_0, y) + \frac{\partial f}{\partial y}(x, y + \theta \Delta y) \cdot \Delta y$$

$$\therefore |f(x, y_2) - f(x, y_1)| = \frac{\partial f}{\partial y}(x, y_1 + \theta(y_2 - y_1)) \cdot |y_2 - y_1|$$

$$\text{而 } \frac{\partial f}{\partial y} = \frac{1}{2\sqrt{y}}, \text{ 在区间 } [0, b] \text{ 上无界}$$

故不满足利普希茨条件

$$3. f(x, y_1) - f(x, y_2) = x^2 y_1 - x^2 y_2 = x^2 (y_1 - y_2)$$

$$\therefore |f(x, y_1) - f(x, y_2)| = x^2 |y_1 - y_2| \leq |y_1 - y_2|, \text{ 故 } L = 1$$

$$f(x, y) \leq f(1, 1) = 2, M = 2.$$

$$h = \min \left\{ 1, \frac{1}{2} \right\} = \frac{1}{2}$$

$$2h = 1$$

2. 设 $\varphi_1(x), \varphi_2(x)$ 是微分方程

$$y'' + q(x)y = 0$$

的任意两个解, 其中 $q(x)$ 在 (a, b) 上连续. 证明: $\varphi_1(x)$ 与 $\varphi_2(x)$ 的朗斯基行列式

$$W(x) \equiv \text{常数}, \quad x \in (a, b).$$

3. 设 $y = \varphi(x)$ 是线性齐次微分方程 (9.61) 的一个非零解 (即 $\varphi(x) \neq 0, x \in (a, b)$), 若有 $x_0 \in (a, b)$ 使 $\varphi(x_0) = 0$, 证明: $\varphi'(x_0) \neq 0$.

4. 设 $\varphi_1(x)$ 与 $\varphi_2(x)$ 是线性齐次微分方程 (9.61) 的两个 线性无关的解. 证明: 它们 没有公共的零点.

$$2. \quad W(x) = \begin{vmatrix} \varphi_1(x) & \varphi_2(x) \\ \varphi_1'(x) & \varphi_2'(x) \end{vmatrix} = \varphi_1(x) \varphi_2'(x) - \varphi_1'(x) \varphi_2(x)$$

$$\begin{cases} \varphi_1''(x) = -q(x)\varphi_1(x) \\ \varphi_2''(x) = -q(x)\varphi_2(x) \end{cases}$$

$$\text{则} \quad \varphi_1''(x) \cdot \varphi_2(x) = \varphi_2''(x) \varphi_1(x)$$

$$\textcircled{1} \text{ 若 } q(x) = 0, \quad y'' = 0, \quad y' = C_1, \quad y = C_1 x + C_2$$

$$\begin{aligned} W(x) &= (C_1 x + C_2) C_1' - C_1 (C_1' x + C_2') \\ &= C_2 C_1' - C_1 C_2', \text{ 为常数} \end{aligned}$$

$$\textcircled{2} \text{ 若 } q(x) \neq 0, \quad \varphi_1''(x) \varphi_2(x) = \varphi_2''(x) \varphi_1(x)$$

$$\int \varphi_1''(x) \varphi_2(x) dx = \int \varphi_2(x) d\varphi_1'(x) = \varphi_2(x) \varphi_1'(x) - \int \varphi_1'(x) \varphi_2'(x) dx$$

$$\int \varphi_2''(x) \varphi_1(x) dx = \int \varphi_1(x) d\varphi_2'(x) = \varphi_1(x) \varphi_2'(x) - \int \varphi_1'(x) \varphi_2'(x) dx$$

$$\text{同时积分, 则有 } \varphi_2(x) \varphi_1'(x) = \varphi_1(x) \varphi_2'(x) + C$$

$$\text{即} \quad W(x) \equiv \text{常数 } C$$

$$4. \quad y''(x) + p(x)y'(x) + q(x)y(x) = 0, \text{ 由于线性无关, 则有 } W(x) \neq 0, \quad \varphi_1(x) \varphi_2'(x) - \varphi_1'(x) \varphi_2(x) \neq 0$$

$$\text{若存在某点 } x_0 \text{ 使 } \varphi_1(x_0) = \varphi_2(x_0) = 0, \text{ 则有 } W(x_0) = 0, \text{ 矛盾}$$

故不存在公共零点

